Causal Source–Report Alignment (CSRA) framework

A. Controlled factual item

Question q_i

Three answer roles:

- P: parametric answer (no-document preference)
- C: conflicting contextual answer
- I: irrelevant alternative

Constructs separated:

- answer selection
- source influence
- verbal attribution

B. Matched document interventions

For source strength s in {skeptical, provisional, bare}:

$D_{P,s}$: document asserts P

$D_{C,s}$: matched document asserts C

Control condition:

$D_{I,\text{provisional}}$ as irrelevant-context control

Matched on source-strength framing; claim content varies between P and C.

C. Behavioral source influence

Permutation-averaged log supports:

$$\bar{l}^{(m)}_{i,d}(P),\ \bar{l}^{(m)}_{i,d}(C),\ \bar{l}^{(m)}_{i,d}(I)$$

Relative contextual support:

$$R^{(m)}_{i,d} = \bar{l}^{(m)}_{i,d}(C) - \bar{l}^{(m)}_{i,d}(P)$$

Causal influence estimate (CIE):

$$\text{CIE}^{(m)}_{i,s} = R^{(m)}_{i,D_{C,s}} - R^{(m)}_{i,D_{P,s}}$$

Positive CIE indicates a shift in relative support toward contextual answer C.

D. Verbal source reports

After the answer is frozen:

SELF and PEER prompts use the same question, document, and recorded answer.

Report roles:

document prior both uncertain

Verbal source-attribution score (VSA):

$$\text{VSA}^{(m)}_{i,d,f} = \ell^{(m)}_{i,d,f}(\text{document}) - \ell^{(m)}_{i,d,f}(\text{prior})$$

E. Validation analyses

SELF–PEER alignment:

Δ_m: difference between SELF and PEER standardized CIE slopes for standardized VSA

Cross-family contrast:

$$H = \Delta_{\text{Qwen}} - \frac{\Delta_{\text{Granite}} + \Delta_{\text{Phi}}}{2}$$

Incremental predictive validation:

X_1: visible task features
X_2: X_1 + PEER VSA
X_3: X_2 + SELF VSA

Diagnostics:

alternate wording and fictional controls

Notation: q_i = question; P, C, I = semantic answer roles; $D_{r,s}$ = document asserting role r at source strength s; CIE = causal influence estimate; VSA = verbal source-attribution score.